



DEPARTMENT OF

COMPUTER ENGINEERING

UNIVERSITY OF ILORIN, ILORIN

STUDENTS' HANDBOOK
(UNDERGRADUATE)

OCTOBER 2018

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FOREWORD

This is the third edition of the Undergraduate Students' Handbook of the Department of Computer Engineering, University of Ilorin. The Department of Computer Engineering through this handbook provides every student in the Department with information that will guide them on issues relating to academic activities in the Department. Information on the Department academic programme, registration of courses and grading system are contained in this handbook. Students will also find useful information on admission and graduation requirements in the Department of Computer Engineering.

Being a new Department running a programme in a highly dynamic field of study, we will keep updating information in this handbook to better serve our students, the University community and anyone who may be interested in getting accurate information about academic activities in the Department of Computer Engineering, University of Ilorin.

For other information regarding conduct of students on campus and support services provided by the University of Ilorin, students are encouraged to consult the University of Ilorin Students' Handbook.

We wish all our students the very best in their academic pursuit.

Engr. Dr. J.F. Opadiji
(Ag. Head of Department)

STAFF LIST

J. F. Opadiji	B.Eng, M.Eng (Ilorin), Dr.Eng. (Kobe), MNSE, R.Engr.	Senior Lecturer & Ag. Head
A.T. Ajiboye	B.Sc. (Ibadan), M.Eng., Ph.D. (Ilorin), MNSE, R.Engr.	Senior Lecturer
S. A. Y. Amuda	B.Eng. (Yola), M.Eng., Ph.D. (Ilorin), MNSE, R.Engr.	Lecturer I
A.F. Salami	B.S. (Malaysia), M.S. (Malaysia)	Lecturer II
S.A. Olatunji	B.Eng., M.Eng. (Ilorin) MNSE, R.Engr.	Lecturer II
S.L. Ayinla	B.Eng., M.Eng. (Awka) MNSE, R.Engr.	Lecturer II
A.R.Ajayi	B.Eng (LAUTech) MNSE, R.Engr.	Assistant Lecturer
A.O. Yusuf	B.Eng. (Minna)	Graduate Assistant
H.O. Mahmud	B.Eng. (Ilorin)	Technologist I
O.F. Adebayo	B.Eng. (Ilorin)	Technologist II
R.A. Abdulwahab	SSCE	Laboratory Assistant
Y.J. Ajayi	ND (Bida)	Confidential Sec. II
L.A. Folami	SSCE, B.Sc. (Ilorin)	Assistant Exec. Officer
S. Adebisi	SSCE	Head Office Assistant

HISTORY OF THE PROGRAMME

The Department of Computer Engineering, University of Ilorin was established in August 2014 as an offshoot of the Department of Electrical and Electronics Engineering. The Department admitted her first set of undergraduate students into the Bachelor of Engineering (B.Eng.), Computer Engineering degree programme at the commencement of the 2014/2015 academic session.

The establishment of the Department was aimed at providing a platform for multidisciplinary and cross-disciplinary academic activities aimed at providing solutions to industrial and socioeconomic challenges through training of highly skilled manpower for the increasingly dynamic ICT industry and conducting of cutting-edge research for industrial development.

The curriculum for the Computer Engineering programme was developed by a team of experienced academics, who have painstakingly crafted the academic programme to conform to the NUC and COREN Benchmark Minimum Academic Standards (BMAS) while making provision for students to be well-versed in state-of-the-art computer technologies.

PHILOSOPHY AND OBJECTIVES OF THE PROGRAMME

The philosophy of the Computer Engineering programme is focused on producing graduates who are producers rather than mere consumers and technicians of knowledge economy. After graduation, they become functional computer engineers in industries, scholars in academia, and successful entrepreneurs in all aspects of computer engineering. Therefore, the programme would contribute effectively and sufficiently to the knowledge economy by putting in place curriculum that meets these challenges in Computer Engineering.

The objectives of the programme include, to

- (i) deploy curriculum that covers broad and dynamic computer engineering principles, for various computer skill acquisition;
- (ii) produce graduates who are exposed to extensive and appropriate lab experiments and practical work experience;
- (iii) produce graduates who are equipped in design, implementation, maintenance of computer and embedded systems;
- (iv) produce manpower for private and public sectors of the economy as well as job creation.

LABORATORIES/WORKSHOPS

Starting from the second year, students in the Department are expected to conduct experiments and workshop practical sessions in the following laboratories

S/N	DESCRIPTION
1.	Electrical Laboratory
2.	Electronic Laboratory
3.	Microprocessor Development Laboratory
4.	Software Engineering Laboratory
5.	Embedded Systems & Automation Laboratory
6.	Electrical Energy Systems Laboratory
7.	Computer Maintenance Workshop

Students in the Department are also expected to make use of the University Central Workshop (housed in the Faculty of Engineering and Technology for project fabrication), as well as the University Central Research Laboratory.

ADMISSION REQUIREMENTS

- (i) **Unified Tertiary Matriculation Examinations (UTME) Candidates:**
UTME candidates are required to obtain O' Level credits in five subjects which should include Mathematics, Physics, Chemistry and English Language
- (ii) **Direct Entry (DE) Candidates into 200 Level:**
In addition to (i) above, DE candidates are required to have A' Level or equivalent pass in Mathematics, Physics and Chemistry or a National Diploma (ND) in Computer Engineering or a relevant Engineering discipline, at distinction or Upper Credit level, from a recognized Polytechnic or College of Technology plus at least one year post ND experience.
- (ii) **Direct Entry (DE) Candidates into 300 Level:**
In addition to (i) above, DE candidates in this category are required to possess a Higher National Diploma (HND) in Computer Engineering or a relevant Engineering discipline, at distinction or Upper Credit level, with, at least, a National Diploma (Upper Credit) from a recognized polytechnic or College of Technology.

DURATION OF THE PROGRAMME

The Department operates an academic programme which leads to a Bachelor of Engineering (B.Eng.) degree at a duration of five academic sessions for students who gain admission through UME/UTME; and four academic sessions for students who come in as Direct Entry with A-level/IJMB or approved equivalent.

The five-year programme is broken into nine semesters of residence and two industrial attachments. Each semester is usually fifteen teaching weeks long and

instruction is by courses quantified into course credits. Lectures, tutorials, practical sessions and projects are organized during each semester and every student in the programme is expected to participate fully since the grading system reflects each aspect.

AREAS OF STUDY

The Department runs two options as follows:

- Embedded Systems and Automation
- Computer Hardware and Software Systems

Courses in these options are available at the final (500) level.

INDUSTRIAL TRAINING

The Industrial Training aspect of the degree programme is divided into four parts. The first part is the Student Work Experience Program (SWEP) which is held at the end of the 200-level, just during the end of-session break in the Faculty's workshops and laboratories. During this programme students are exposed to workshop practice and have a "hands-on" experience of practical engineering. This serves as a channel for integrating in-house entrepreneurial development and skill into our programme.

The three other parts expose the students to real-life situations in engineering or related establishments within and outside the country. The Department has recorded impressive growth in the area of academic, administrative, staffing and infrastructures.

REGISTRATION FOR COURSES

Courses for the session are registered for on-line at the beginning of the session. In addition, a student is required to go to his level adviser after the on-line registration in order to complete the registration process. Failure to register before the deadline, as stipulated by the university, may result in loss of that session or voluntary withdrawal from the programme. A student cannot re-register for a course already passed.

REQUIREMENTS TO PROCEED FROM 100 LEVEL TO 200 LEVEL

In order for a student to proceed from 100 Level to 200 Level in the Faculty of Engineering and Technology (FET), the following minimum requirements must be met at the end of 100 Level:

- (a) Satisfy the 9-9-6 requirement, meaning that a student must pass
 - 9 credits of Mathematics
 - 9 credits of Physics, and
 - 6 credits of Chemistry.
- (b) A cumulative grade point average of 2.00.

STANDARD OF TESTS AND EXAMINATIONS

Examination questions are set by the lecturer(s) assigned to teach the courses. It is mandatory for the examination questions in all courses to be internally reviewed and in addition to be externally moderated in case of 500 level courses. Lecturers submit their question papers along with model answers under strict confidential cover to the Head of Department (the departmental Chief Examiner) who, together with the concerned lecturer and his sectional head, reviews each question paper to ensure that it conforms to standard. Once satisfactorily reviewed and corrected, the question papers are processed under strict confidentiality, by the Head of Department, with the assistance of the departmental Examination Officer, either for onward transmission to the external examiner, and kept in safe custody by the Examination Officer, pending a few hours before the time of the examination.

Final year examination questions are particularly moderated by External Examiners who are seasoned, experienced and long-standing professionals in various fields of Computer Engineering. At the end of every session, External Examiners are invited to vet question papers, results, project reports and conduct a project defense for final students in the department.

GRADING SYSTEM

The University operates a 5-point grading system in which the continuous assessment takes 30% and examination 70% in each course for each semester. The Continuous Assessment is made up of class attendance/quiz/test/assignment as applicable. In addition, a student must satisfy a minimum of 70% attendance at lectures before being allowed to sit for examination in the respective course.

The range of scores and the corresponding letter grades are as shown in the Table below.

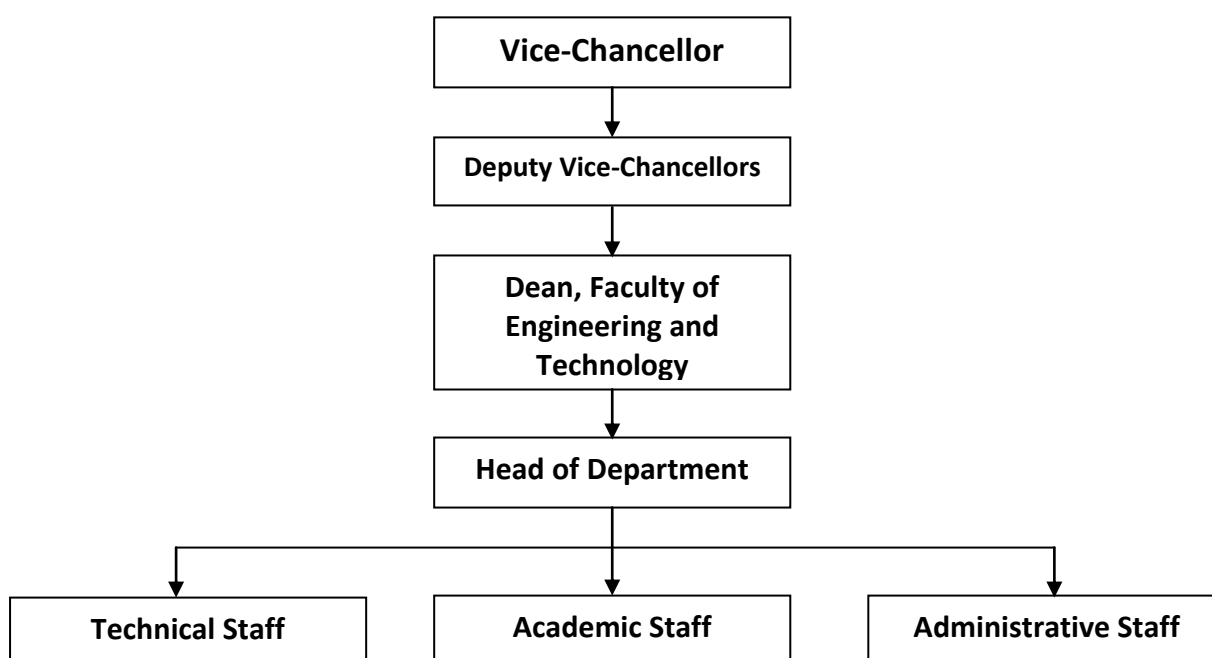
Grade	Grade Point	Range of Scores (%)
A	5.0	70 and above
B	4.0	60 - 69
C	3.0	50 – 59
D	2.0	45 – 49
E	1.0	40 – 44
F	0.0	Less than 40

CLASSIFICATION OF DEGREES

The Bachelor of Engineering (B.Eng.) degree in Computer Engineering is awarded in the First Class, Second Class Upper Division, Second Class Lower Division, and Third Class, depending on the Cumulative Grade Point Average (CGPA) at the point of graduation and as shown in the Table below.

Class of Degree	Range of C.G.P.A.
First Class	4.50 – 5.00
Second Class Upper	3.50 – 4.49
Second Class Lower	2.40 – 3.49
Third Class	1.50 – 2.39

ORGANISATION CHART OF THE UNIVERSITY/DEPARTMENT



PRINCIPAL OFFICERS OF THE UNIVERSITY

Vice- Chancellor: Prof. S.A. Abdulkareem

Deputy Vice-Chancellor (Academic): Prof. Mrs. S.O. Malomo

Deputy Vice-Chancellor (Management Services): Prof. B.F. Sule

Deputy Vice-Chancellor (Research, Training & Innovation): Prof. M.O. Buhari

Registrar: Dr. Mrs. F.M. Olowoleni

University Librarian: Prof. A.O. Issa

Bursar: Mr. S.Y. Abiodun

DEAN, FACULTY OF ENGINEERING AND TECHNOLOGY

Engr. Professor D.S. Ogunniyi (August 2017 – date)

HEADS OF DEPARTMENT OF COMPUTER ENGINEERING:

1. Engr. Dr. J.F. Opadiji (August 2014 – July 2016)
2. Engr. Dr. A.T. Ajiboye (August 2016 – July 2018)
3. Engr. Dr. J.F. Opadiji (August 2018 – date)

ADMINISTRATIVE RESPONSIBILITIES IN THE DEPARTMENT

1. Head of Department
2. Examination Officer
3. Assistant Examination Officer
4. Final Year Project Coordinator
5. Mini-project Coordinator(s)
6. SIWES Representative
7. SWEP Representative
8. COBES Representative
9. 100 Level Adviser
10. 200 Level Adviser
11. 300 Level Adviser
12. 400 Level Adviser
13. 500 Level Adviser
14. Postgraduate Programme Coordinator

COURSE STRUCTURE

100 LEVEL

100 LEVEL HARMATTAN SEMESTER

COURSE CODE	COURSE TITLE	COURSE STATUS	NO. OF CREDIT
CHM 101	General Physical Chemistry I	E	3
CHM 115	General Practical Chemistry I	E	2
GNS 111	Use of English I	R	2
MAT 111	Elementary Mathematics I	E	3
MAT 113	Elementary Mathematics II	E	3
PHY 115	Mechanics and Properties of Matter I	E	2
PHY 125	Heat, Sound and Optics	E	3
PHY 191	Practical Physics I	E	1
STA 131	Introduction to Statistical Inference I	E	2
Total			21

100 LEVEL RAIN SEMESTER

COURSE CODE	COURSE TITLE	COURSE STATUS	NO. OF CREDIT
CHM 112	General Organic Chemistry	E	2
CHM 132	General Inorganic Chemistry	E	2
GNS 112	Use of English II	R	2
GNS 114	Digital Literacy Course	R	1
MAT 112	Elementary Mathematics III	E	3
MAT 114	Elementary Mathematics IV	E	3
PHY 152	Electricity and Magnetism I	E	3
PHY 142	Atomic and Nuclear Physics	E	2
PHY 192	Practical Physics II	E	1
STA 124	Introduction to Probability Distribution	E	2
CHM 116	General Practical Chemistry II	E	1
Total			22

200 LEVEL

200 LEVEL HARMATTAN SEMESTER

COURSE CODE	COURSE TITLE	COURSE STATUS	NO. OF CREDIT
ABE 263	Engineering Mathematics I	R	3
CHE 241	Fundamental of Fluid Mechanics	R	3
CVE 253	Applied Mechanics	R	3
CPE 283	Laboratory Course I	C	2
ELE 201	Applied Electricity I	R	3
ELE 275	Computer Programming I	R	1
GNS 211	Introduction to Humanities	R	2
MEE 217	Engineering Graphics I	R	2
MEE 235	Engineering Technology	R	2
Total			21

200 LEVEL RAIN SEMESTER

COURSE CODE	COURSE TITLE	COURSE STATUS	NO. OF CREDIT
ABE 206	Introduction to engineering disciplines	R	2
CHE 242	Fundamental of Thermodynamics	R	3
CHE 264	Engineering Mathematics II	R	3
CPE 284	Laboratory Course II	C	2
CVE 254	Applied Mechanics II	R	3
ELE 202	Applied Electricity II	R	3
ELE 276	Computer Programming II	R	2
GNS 212	Introduction to Social Sciences	R	2
GSE 202	Graduate Skill Acquisition	R	2
MEE 218	Engineering Graphics II	R	2
MEE 272	Engineering Materials	R	2
			26

200 LEVEL THIRD SEMESTER (LONG VACATION)

COURSE CODE	COURSE TITLE	COURSE STATUS	NO. OF CREDIT
CPE 222	Students Work Experience Programme (SWEP)	C	6
Total			6

300 LEVEL**300 LEVEL HARMATTAN SEMESTER**

COURSE CODE	COURSE TITLE	COURSE STATUS	NO. OF CREDIT
CPE 311	Electric Circuit Theory	C	3
CPE 321	Analogue Electronics Circuits	C	3
CPE 331	Electromagnetic Fields & Waves	C	3
CPE 341	Software Development Techniques	C	3
CPE 381	Laboratory Course I	C	1
GNS 311	History and Philosophy of Science	R	2
GSE 301	Entrepreneurial Skill	R	2
MEE 361	Engineering Mathematics III	R	3
Total			20

300 LEVEL RAIN SEMESTER

COURSE CODE	COURSE TITLE	COURSE STATUS	NO. OF CREDIT
ABE 306	Engineering Economics	R	2
ABE 376	Engineering Communication	R	1
CPE 312	Measurement & Instrumentation	C	2
CPE 336	Digital Electronics	C	3
CPE 342	Software Engineering	C	3
CPE 372	Data Communication & Networks	C	3
CPE 382	Laboratory Course & Mini-Project	C	2
MEE 362	Engineering Mathematics IV	R	3
Total			19

300 LEVEL THIRD SEMESTER (LONG VACATION)

COURSE CODE	COURSE TITLE	COURSE STATUS	NO. OF CREDIT
CPE 392	Industrial Training I	C	6
Total			6

400 LEVEL**400 LEVEL HARMATTAN SEMESTER**

COURSE CODE	COURSE TITLE	COURSE STATUS	NO. OF CREDIT
CPE 423	Introduction to Digital System Design with VHDL	C	2
CPE 435	Prototyping Techniques & Packaging	C	2
CPE 437	Artificial Intelligence and Applications	C	2
CPE 443	Assembly Language Programming	C	2
CPE 441	Control Theory I	C	3
CPE 451	Communication Principles	C	2
ABE 463	Engineering Statistics	R	2
CPE 471	Microprocessor System and Interfacing	C	3
CPE 481	Laboratory Course III	C	2
Total			20

400 LEVEL RAIN SEMESTER

COURSE CODE	COURSE TITLE	COURSE STATUS	NO. OF CREDIT
CPE 492	Industrial Training II	C	12
Total			12

500 LEVEL**500 LEVEL HARMATTAN SEMESTER****COMMON COURSES:**

COURSE CODE	COURSE TITLE	COURSE STATUS	NO. OF CREDIT
ABE 573	Engineer in Society	R	1
CPE 501	Design & Installation of Electrical & ICT services	C	2
CPE 531	Digital Signal Processing	C	3
CPE 541	Control Theory II	C	2
CPE 543	Technopreneurship & Cyberlaw	C	2
CPE 561	Computer Organisation & Architecture	C	2
CPE 593	Project I	C	4
ABE 501	Engineering Management	R	3
Total			19

ELECTIVE COURSES

Embedded Systems & Automation

COURSE CODE	COURSE TITLE	COURSE STATUS	NO. OF CREDIT
CPE547	Power Electronics	E	2
CPE 545	Control Theory III	E	2
Total			4

Computer Hardware & Software Systems

COURSE CODE	COURSE TITLE	COURSE STATUS	NO. OF CREDIT
*CPE 551	Parallel Processing	E	2
*CPE 553	Digital Image Processing	E	2
*CPE 557	Digital Speech Processing	E	2
Total			4

* Students in this option are required to offer at least any of the two elective courses.

500 LEVEL RAIN SEMESTER

COMMON COURSES:

COURSE CODE	COURSE TITLE	COURSE STATUS	NO. OF CREDIT
BUL 506	Engineering Law	R	3
CPE 502	Reliability & Maintainability of Computer system	C	2
CPE 532	Computer Security Techniques	C	2
CPE 546	Embedded System Design & Programming	C	2
CPE 594	Project II	C	4
MME 524	Materials Selection and Economics	R	3
Total			16

ELECTIVE COURSES

Embedded Systems & Automation

COURSE CODE	COURSE TITLE	COURSE STATUS	NO. OF CREDIT
CPE 544	Introduction to Robotics & Automation	E	2
CPE 548	Fuzzy Logic and Programming	E	2
Total			4

Computer Hardware & Software Systems

COURSE CODE	COURSE TITLE	COURSE STATUS	NO. OF CREDIT
CPE 552	Advanced Digital Design	E	2
CPE 554	Cryptography Principles and Applications	E	2
CPE 556	Multimedia Technology & Programming	E	2
Total			4

* Students in this option are required to offer at least any of the two elective courses.

COURSE SYNOPSIS

DETAILS OF TAUGHT COURSES FOR B.ENG DEGREE PROGRAMME IN COMPUTER ENGINEERING

100 LEVEL COURSES

CHM 101 General Physical Chemistry**3 Credits**

Atom, Dalton's atomic theory, atomic masses, Fundamental particles of the atom, atomic structure, Modern electronic theory of atoms. Periodicity of the elements, Mole concept, chemical formulas, equation and calculations, State of matter: gas, liquid and solid, Energetic and thermo chemistry, Chemical kinetics, equilibriums and electrochemistry. 45h(T); E

CHM 112 General Organic Chemistry**2 Credits**

Historical survey of the development and importance of organic Chemistry Nomenclature and classes of organic compounds, Homologous Series, functional groups, isolation and purification of organic compounds. Qualitative and quantitative organic chemistry, stereochemistry, saturated hydrocarbons and unsaturated hydrocarbons. 30h (T); E

CHM 115 General Practical Chemistry I**2 Credits**

Theory and practice of quantitative chemical analysis, Acid-base, oxidation- reduction reactions, precipitation and complexometric titrations. Gravimetric analysis. Calculation, data analysis and presentation. 15h (T), 45h (P); E

CHM 116 General Practical Chemistry II**1 Credit**

Qualitative inorganic and organic analysis for elements in group IA, IIA, IIIA, IVA, IB, IIB and IIIB. Chemical analysis for functional groups: acidic, ketonic, carboxylic etc. 45h (P); E

CHM 132 General Inorganic Chemistry**2 Credits**

Periodic tables and periodic properties, chemical bonding, Structures of solids, the chemistry of selected representative elements, Qualitative analysis, hybridization. 30h (T); E

GNS 111 Use of English I**2 Credits**

Information collection and usage. Collection and organization of materials. Logical presentation of papers. Use of library. Reference sources. Techniques of fast reading and writing. Continuous writing. Oral communication, Public speaking and phonetics. 15h (T); 45h (P); R

GNS 112 Use of English**2 Credits**

A basic course in effective listening skills. Use of the English Language relevant to students' disciplines. Structure of the sentence types. Verbs. Classification of nouns, punctuation. Writing of essays. Sentence Construction. Outlines and paragraphs. 15h (T); 45h (P); R

MAT 111 Elementary Mathematics I**3 Credits**

Elementary set theory, subsets, union, intersection, complements, venn diagrams. Real numbers, integers, rational and irrational numbers. Mathematical induction, real sequences and series, theory of Quadratic equation, Binomial theorem, complex numbers, algebra of

complex numbers, argand diagram, de Moivre's theorem, n-th roots of unity, Circular measure, trigonometric functions of angles of any magnitude, addition and factor formulae.

30h (T); POT 15hrs; E.

MAT 112 Elementary Mathematics III

3 Credits

Functions of a real variable, graphs, limits and idea of continuity. The derivative, as limit of rate of change. Techniques of differentiation, maxima and minima. Extreme curve sketching, integration Definite integrals, reduction formulae, application to areas, volumes (including approximate integration: trapezium and Simpson's rule).

30h (T) PR: MAT 101,102, (L.30P.O.T.15); E

MAT 113 Elementary Mathematics II

3 Credits

Geometric representation of vectors in 1-3 dimensions, components, Direction cosines. Addition, scalar, multiplication of vectors, linear independence, Scalar and vector products of two vectors. Differentiation and integration of vector with respect to a scalar variable. Two dimensional coordinate geometry. Straight lines, Circles, parabola, ellipse, hyperbola. Tangents, normals. Kinematics of a particle. Component of velocity and acceleration of a particle moving in a plane. Force momentum ,laws of motion under gravity, projectiles, resisted vertical motion, elastic string, simple pendulum, impulse ,impact of two smooth spheres and of a sphere on a smooth surface.

30h (T),45h (P),C

MAT 114 Elementary Mathematics IV

3 Credits

Mapping, bisection, composition ,inverse mapping, binary operations, associativity, identity elements, invertible elements, distributivity. Relations, fundamental theorem of equivalence relations, Trigonometric ratio, sums and product formulae. Multiple and sub-multiple angles. Graphs of trigonometric functions. Inverse circular functions. Solution of triangles and trigonometric equations. Height and distance in 2 and 3 dimensions. Geometrical equation of lines and planes and other applications. Angle between two lines. Taylor's and Maclaurin's theorems. Simple examples. 30h(T), 45(P), PR: MAT 101; E.

PHY 115 Mechanics and Properties Of Matter I

2 Credits

Units and dimensions, scalar and vectors, particle kinematic ,Newton's law, friction,work, energy, centre of mass, simple harmonic motion,rigid body dynamics,Kepler's laws, pressure in fluids,intermolecular forces, Hooke's law, young's modulus,fluid flow streamline turbulence, Stoke's laws,surface tension .

30h (T); E

PHY 125 Heat, Sound And Optics

3 Credits

Temperature, thermometers, heat transfer pVT surfaces, kinetic Theory, first and second laws of the thermodynamics transverse and longitudinal waves, standing waves,intensity beats, Doppler effect, electromagnetic spectrum, Huygen's principles,images formed by a single surfaces,thin lenses,aberrations, the eye,optical instruments, interference, single slit iffraction,diffraction grating, polarization, Malus's law.

45h (T); E

PHY142 Atomic and Nuclear Physics

2 Credits

Theory of atomic structure, Thompson, Rutherford and Bohr's theories, the hydrogen atom. Properties of the electron, e/m, C.R.O., Milikan's experiment,properties of the nucleus, natural radioactivity, wave particles duality of light, X-rays,photoelectricity, Thermionic emission, Diode valve.

30h (T); E

PHY 152 Electricity and Magnetism I

3 Credits

Coulombs's law, Gauss theorem, capacitors, Ohm's law,Kirchoff's law electrical energy, D.c bridges, potentiometer, magnetic effect of current, electromagnetic induction, moving coil

and ballistic galvanometer, multimeters, d.c and a.c motors and generators, Hysteresis, power in A.C circuit, semiconductor, conductivity and mobility rectification.

45h (T); E

PHY 191 Practical Physics I

1 Credit

A selection from the following experiment, use of measuring instrument, viscosity, surface tension, oscillations about an equilibrium position, Hooke's law, moment of inertia, focal length of lens, active index, optical instruments, the sonometer, heat capacity, volume expansion and latent heat.

45h (P); E

PHY 192 Practical Physics II

1 Credit

A selection from the following experiments. Potential difference and internal resistance of cells, uses of potentiometer circuit, the metre bridge, simple direct current measuring instruments, Planck's constant, radioactivity.

45h (P); E

STA 124 Introduction to Probability Distribution

2 Credits

Random variable, Bernoulli trials, Binomial, Geometric, Poisson, Uniform and normal distributions. Bivariate. Marginal and condition distributions, Concepts of linear regression & Correlation and association of attributes.

30h (T); E

STA 131 Introduction to Statistical Inference I

2 Credits

Statistical data: their source, collection and preliminary analysis by table, graphs and simple statistics to include measures of location dispersion, skewness, kurtosis and correlation.

30h (T).

200 LEVEL COURSES

ABE 263 Engineering Mathematics I

3 Credits

Limits, Continuity, differentiation, introduction to linear first order differential equations, partial and total derivatives of composite functions, matrices and determinants, Vector algebra, Vector calculus, Directional Derivatives.

45h (T); R

CHE 241 Fundamentals of Fluid Mechanics

3 Credits

Properties of fluids, Fluids statics, Basic conservation laws, friction effects and losses in laminar and turbulent flows in ducts and pipes. Dimensional analysis and dynamic similitude, principles of construction and operation of selected hydraulic machinery. Hydropower systems.

45h (T); R

CHE 242 Fundamentals of Thermodynamics

3 Credits

Basic concepts, quantitative relations of Zeroth, first, second and third laws of thermodynamics. Behaviours of pure substances and perfect gases. Ideal gas cycles.

45h (T); R

CHE 264 Engineering Mathematics II

3 Credits

Second order differential equations, line integral, multiple integral and their applications, differentiation of integral. Analytical functions of complex variables. Transformation and mapping. Special function.

45h (T); PR; MAT 112; R

CPE 222 Students Work Experience Programme**6 Credits**

Introduction to practices and skills in general engineering through instruction in operation of hand and powered tools for wood and metal cutting and fabrication. Supervised hands – on experience in safe usage of tools and machines for selected tasks.

270h (P); C

CPE 283 General Engineering Laboratory Course I**2 Credits**

Laboratory investigation and report submission for selected experiments and projects in Applied Mechanics, Applied Electricity I and Fundamentals of Fluid Mechanics.

90h (P); C

CPE 284 General Engineering Lab Course II**2 Credits**

Laboratory investigations and report submission for selected experiments and Projects in fundamentals of Thermodynamics. Engineering Materials, Applied Mechanics II and Applied Electricity II.

90h (P); C

CVE 253 Applied Mechanics I**3 Credits**

Forces, moments, couples. Equilibrium of simple structures and machine parts. Friction. First and second moments of area; centroids. Kinematics of particles and rigid bodies in plane motion. Newton's laws of motion. Kinetic energy and momentum analyses.

45h (T); R

CVE 254 Applied Mechanics II**3 Credits**

Hooke's law; stresses and strains due to loading and temperature changes. Torsion. The stress circle, deflection of beams with symmetrical and combined loading. Elastic buckling of columns, Shear forces and bending moments, analytical and graphical methods for structures.

45h (T); R

ELE 201 Applied Electricity I**3 Credits**

Fundamental concepts – Electric Fields, charges, magnetic fields. Current, B.—H Curves, Kirchoff's laws, superposition. Thevenin, Norton theorems, Reciprocity, RL, RC, RLC circuits. DC bridges. Resistance, Capacitance and Inductance measurements. Transducers. Single phase circuits, Complex J – notation , AC circuits, impedance, admittance, susceptance.

45h (T); R

ELE 202 Applied Electricity II**3 Credits**

Basic machines – DC, synchronous alternators, transformers, equivalent circuits. Three phase balanced circuits, PN junction Diode, Transistors, Thyristors, FETs, Zener, Rectifiers. Basic control systems, open/closed loop systems. Communications fundamentals, introduction of TV,, Radio, Telephone systems.

45h (T) PR; ELE 201; R

ELE 275 Computer Programming I**1 Credit**

Identification, functions, applications, and use of PC parts and peripheral devices. Safety precautions and preventive maintenance of PC. Filing system: Word processing applications and use. Internet: available services, principle of operation, applications, demonstrations. .Spreadsheet: applications, and how to use. Database Management package: applications, demonstrations. Report Presentation Software Packages: applications, demonstrations, and use. Mini-project to test proficiency in use of these software packages.

15h (T), R

ELE 276 Computer Programming II**2 Credits**

Program design using pseudo-code/flowchart. Extensive examples and exercises in solving engineering problems using pseudo-code/flowchart. Computer programming using structure BASIC such as QBASIC: symbols, keywords, identifiers, data types, operators, statements, flow of control, arrays, and functions. Extensive examples and exercises in solving engineering problems using QBASIC. Use of Visual programming such as Visual BASIC in solving engineering problems.

30h (T), R

GNS 211 Philosophy, Logic and Nigerian Culture**2 Credits**

Communication in human society. Human creativity. Foundations of philosophy. Freedom, human rights and concept of justice. Logical Thought and critical reasoning. Laws of thought. African civilizations, development, religions and culture. Nigerian perception of his world. Culture areas of Nigeria and their characteristics.

30h (T); R

GNS 212 Introduction to Social Sciences and Citizenship Education**2 Credits**

Basic Principles of Sociology and Anthropology. The development process. The Nigerian Economy –History and Policies. Citizenship Education in Nigeria- Constitution, Arms of the Nigerian Government, Right and Obligations, National Ethics and Discipline in National Life, National identity, Nigerian Environment. Foreign Policy and International Relations. Social ills-Examination malpractice, bankruptcy etc., and their sanctions. Basic principles of accounting and management.

30h (T); R

MEE 217 Engineering Graphics I**2 Credits**

Lettering, Dimensioning, Orthographic projection, auxiliary and sectional views, True lengths, Graphical calculus and Architectural Drawings.

15h (T), 45h (P); R

MEE 218 Engineering Graphics II**2 Credits**

Advanced topics in auxiliary and sectional views, development, intersection of surfaces, isometric projection; dimensioning and tolerances; Blue – print reading.

15h (T); 45h (P); R

MEE 235 Engineering Technology**2 Credits**

Introduction to workshop practice. Industrial safety, machine tools and fabrication technology. Use of hand tools and Engineering approach to design.

15h (T); 45h (P); R

MEE 272 Engineering Materials.**2 Credits**

Introduction to electronic configuration, atomic structures, inter-atomic bonding mechanisms, crystal and microstructure. Relationship between structure and properties of metals, alloys, ceramics and plastics. Principles of the behavior of materials in common environments. Fabrication processes and applications.

30h (T); R

300 LEVEL COURSES**ABE 306 Engineering Economics****2 Credits**

The nature and scope of economics, Basic concepts of engineering economy, Interest formulae. Discounted cash flow: Present worth, equivalent annual growth and rate of return comparisons. Replacement Analysis, Breakdown Analysis, Benefit-cost analysis. Minimum acceptable rate of return. Judging attractiveness of proposed investments.

30h (T); R

ABE 376 Engineering Communication**1 Credit**

Professional use of English Language of letters, specification descriptions, presentation of charts, graphs, tables, writing of proposals in reports. Case studies of major professional presentation of reports and proposals.

15h (T); PR: GNS 112; R

CPE 311 Electric Circuit Theory**3 Credits**

Electric quantities: Voltage and current sources, resistor, inductor and capacitor. Circuit laws and theorems. Signals and waveforms. Alternating current behavior in R, L and C circuits. Phasor analysis with complex algebra, Two terminal networks - impedance, admittance and their real and imaginary parts. Resonance: series and parallel resonance, half power points, bandwidth, Q-factors. Power Quantities. First order LR and RC circuits. Complementary function and particular integral. Network topology. Nodal and mesh network analysis. Impedance transformations, Norton's transformation, Bartlett theorem. CAD circuit analysis and simulation.

45h (T); PR: ELE201; C

CPE312 Measurement and Instrumentation**2 Credits**

General Instrumentation, Basic Meter in DC measurement, Basic meter in AC measurements; DC and AC bridges and their applications; Electronic instruments for the measurement of voltage, current resistance and other circuit parameter, electronic voltmeters, AC voltmeters using rectifiers, electronic multimeter, digital voltmeters; oscilloscope: vertical deflection system, horizontal deflection system, probes, sampling CRO; square-wave and pulse generator, signal generators, function generators, wave analysers, Electronic counters and their applications: time base circuitry, universal counter measurement modes; Analog and digital data acquisition systems.

30h (T); C

CPE 321 Analogue Electronic Circuits**3 Credits**

Atom, Energy-band, electrons and holes, P and N doping, impurity level, Transport phenomena in semiconductor. Open-circuited p-n junction, diode currents, V-I characteristics, ideal diode, diode resistance, piecewise linear model, breakdown diodes, diode capacitance. **Field Effect Transistor:** JFET, MOSFET, Biasing, Small-signal model, FET amplifier analysis, FET as voltage-variable resistor. Transistor current components, Common-base configuration, Common-emitter configuration, Common-collector configuration, Q point, Fixed bias, Emitter bias, bias stability, thermal stability. Analysis of CE configuration, Cascading transistor amplifiers, Miller's Theorem. Half-wave rectifier, full-wave rectifier, choke and capacitor filters, peak detector, clipping circuits, Pulse shaping, clamping circuits, voltage doubler.

45h (T); C

CPE 331 Electromagnetic Fields & Waves**3 Credits**

Review of Vector Algebra & Calculus. Gauss's, Stokes, Hemholtz and Green's integral theorems, integral of scalar and vector fields. **Electrostatics.** Coulomb's Law. Concept of fields. Electric flux density and electric field intensity. Gauss's Theorem and applications. Voltage and electric potential. Conductor, dielectrics. Polarization, susceptibility, permittivity. Electrostatic boundary condition. Capacitance calculation and electric energy. **Magnetostatics.** Magnetic flux density and magnetic field intensity. Biot-Savart Law and Ampere's Law, Faraday's Law. Self and mutual induction. Inductance calculation and magnetic energy. **Magnetic Circuits:** B and H, Magnetic materials. Saturation and hysteresis, Hysteresis, eddy current loss, analysis of linear magnetic circuits. **Maxwell's Equations.** Faraday's Law and Lorentz force law. Uniform plane waves and wave equation. Time harmonic fields. Polarization of waves. Poynting's Theorem and the conservation of energy, the field definitions of impedance, admittance. **Wave Propagation and Transmission Theory.** 45h (T); C

CPE 336 Digital Electronics**3 Credits**

Boolean algebra, theorems, switching functions, truth tables, derivation of canonical forms. Switching circuits. Electronic logic gates, basic functional components. Analysis of combinational circuits. Synthesis of combinational logic circuits. Characteristics of minimization methods. Karnaugh maps. Quine-McCluskey tabular minimization method. Computer-aided minimization of switching functions. Algebraic methods for determining prime implicants. Digital vs. analog systems. Mixed signal design, analogue and digital grounding. Digital system design hierarchy. Logic devices: TTL and CMOS families. Interference and noise. **Number Systems and Codes**. Memory devices. Latches: set-reset latch, gated SR latch, delay latch. Flip-flops. Timing circuits. Modular Design. Decoders and Encoders.

45h (T); C

CPE 341 Software development Techniques**3 Credits**

Software development life cycle. Top-down design. Program design using pseudo-code, flowchart. Flowchart ANSI symbols and usage. Extensive examples, and exercises using pseudo-code/flowchart to solve practical problems in engineering. Debugging and documentation techniques. Programming using a structured language such as C: Symbols, keywords, identifiers, data types, operators, various statements, operator precedence, type conversion, conditional and control structures, function, recursive functions. Parameters passing in functions. Arrays. Pointers. Structure, union and enumerated types. Dynamic memory allocation. File Handling: Concept of a file, files and streams, standard file handling functions, binary files, random access files. Advanced Topics: Command line parameters, pointers to functions, creation of header files, stacks, linked lists, bitwise manipulation.

45h (T); 45h (P); PR: ELE 276; C

CPE 342 Software Engineering**3 Credits**

Introduction: Principles of software engineering. Software life cycle. Project management. Computer based system engineering. Requirements and specification: Analysis, definition, specification, software prototyping, formal specification, algebraic specification and mode-based specification. Software Design Architectural design. Object-oriented design. Function-oriented design. Real-time system design. User interface design. Dependable Systems. Reliability and reusability. Safety-critical consideration. Good programming practice. Computer-aided Software Engineering (CASE). Verification and Validation: Validation and testing. Problems of assessing and quantifying the system reliability. Test case and test data design. Management: People and organization issues. Cost estimation. Quality management. Process improvement. Maintenance, Configuration and re-engineering of software.

30h (T); 45h (P); PR: ELE 276, CPE 341; C

CPE 372 Data Communications and Computer Network**2 Credits**

Introduction to Data Communication: Transmission definitions, transmission codes, transmission modes, parallel transmission, serial transmission, bit synchronization, character synchronization, synchronous transmission, asynchronous transmission, efficiency of transmission. error detection methods and data compression. Introduction to network protocol. Seven layer ISO-OSI standard protocols and network architecture. Local Area Networks Distributed system, PBX and cable based LANs. Topology, Medium access control methods. High speed and bridged local area networks. Cyclic redundancy check (CRC). Network operating system. Wide Area Networks. Internet architecture and protocols. LAN network-control centre.

30h (T); C

CPE 381 Laboratory Course I**1 Credit**

Laboratory experiments drawn from all engineering courses offered in Harmattan semester.

135h (P); C

CPE 382 Laboratory Course & Mini-Project**2 Credits**

Laboratory experiments drawn from all the engineering courses offered in Rain semester. Mini-projects to illustrate understanding of the courses.

90h (P); C

CPE 392 SIWES: Industrial Training I**6 Credits**

On the job experience in industry chosen for its relevance to student's major. (12 weeks during the long vacation following 300 level).

270h (P); C

GNS 311 History and Philosophy of Science**2 Credits**

Scientific Methods. Scientific Revolution and Methodology. African Physical Environment. Agriculture in Nigeria. Man and the Universe- His origin, nature and cosmic environment. Concept of disease. Parasites and diseases. Use and abuse of human body. Concept of matter. Energy Sources. Generation distribution, interconversion and conversion of energy. Threat and implication of nuclear war. Science and Technology in the Society and Service of man. Relevance of Scientific Research and advances in human society. Ethics and Technology. Environment effects of chemicals, plastics, textiles, and wastes. Environmental Sanitation. Chemical and radiochemical hazards. Computer and its applications.

30h (T); R

MEE 361 Engineering Mathematics III**3 Credits**

Numerical analysis and its application to engineering problems. Operational methods transform, series and special functions in engineering.

45h (T); PR ABE 263; R

MEE 362 Engineering Mathematics IV**3 Credits**

Mathematical modeling of physical systems, numerical techniques, boundary value problems, Fourier integral, Fourier series, orthogonal functions and Sturm-Liouville systems. Partial differential equations including theory, classification and solution by various methods.

45h (T); PR: CHE 264; R

400 LEVEL COURSES**ABE 463 Engineering Statistics****2 Credits**

Descriptive Statistics, frequency distribution, populations and sample, central tendency, variance data sampling, mean, median, mode, mean deviation, percentiles etc. Probability, Binomial, Poisson, hyper geometric, normal distributions, etc. Statistical inference intervals, tests, hypothesis and significance. Regression and Correlation.

30h (T); R

CPE 423 Introduction to Digital System Design with VHDL**2 Credits**

Finite State Machine: definition, mealy and moore models, state diagram, state table, transition table. Sequential circuits design using flip-flops, asynchronous, and synchronous circuit design. Algorithm State Machine. Design examples and exercises. Structured Design: Design constructs, Design Levels, Geometry-based interchange formats, Computer aided electronic system design tools, Schematic circuit capture, Hardware description languages, Design process (simulation, synthesis), Structural design decomposition. Introduction to VHDL: VHDL language abstractions, Design hierarchies, VHDL component, Lexical description, VHDL source file, Data types, Data objects, Language statements, Concurrent VHDL, Sequential VHDL.

30h (T); 30h (P); PR: CPE 336; C

CPE 435 Prototyping Techniques & Packaging**2 Credits**

Introduction: Grounding, ground plane, digital ground, analogue ground, power decoupling, inductance and capacitive effects, feedthrough capacitors. Soldering techniques for pass-through and surface mount components, desoldering. Breadboarding, veroboarding. Wire wrapping techniques. Radio Frequency design and implementation techniques. Printed Circuit Board techniques, and production of PCB. Use of PCB CAD packages. Construction exercises using different prototyping techniques. Packaging techniques. Documentation and Manual writing.

30h (T); 30h (P); C

CPE 437 Artificial Intelligence & Application**2 Credits**

Artificial Neural Networks (ANN), Genetic Algorithm (G.A.) Concepts, Simulated Annealing, theories and Applications; Agent-based Systems, supervised and unsupervised learning; reinforcement learning.

30h (T); C

CPE 441 Control Theory I**3 Credits**

Introduction: definition, examples of control systems. Open-loop and closed-loop control systems. Review of Laplace and inverse Laplace transforms. **System modeling:** Signal flow graph, block diagram. Transfer function. Poles and zeros. Block diagram reduction using signal flow graph and block diagram reduction techniques. Mechanical, electrical and electromechanical systems. First and second order models, higher order models. Definitions of transient response parameters. Analysis of second-order system as prototype. Routh-Hurwitz stability criterion. Classification of systems based on steady-state characteristics, steady-state error coefficient.

45h (T); PR: ELE 311; C.

CPE 443 Assembly Language Programming**2 Credits**

Introduction: Language level of abstraction and effect on machine, characteristics of machine code, advantages, justifications of machine code programming, instruction set and dependency on underlying processor. Intel 8086 microprocessor assembly language programming: Programming model as resources available to programmer, addressing modes, instruction format, instruction set- arithmetic, logical, string, branching, program control, machine control, input/output, etc; assembler directives, hand-assembling, additional 80x86/Pentium instructions. Modular programming. Interrupt and service routine. Interfacing of assembly language to C. Intel 80x87 floating point programming. Introduction to MMX and SSE programming. Motorola 680x0 assembly language programming. Extensive practical engineering problems solving in assembly language using MASM for Intel, and cross-assembler for Motorola.

30h (T); PR: CPE 341, CPE 336; C

CPE 451 Communication Principles**2 Credits**

Principles and historical development of Communications. Block diagram of a communication system. The frequency spectrum. Signals and vectors, orthogonal functions, Fourier series, Fourier integral, signal spectrum, convolution, power and energy, correlation. Reasons for modulation. Types of modulation. Amplitude modulation systems. Vestigial sideband. Frequency mixing and multiplying, frequency division multiplexing, applications of AM systems. Frequency modulation systems. Frequency deviation, modulation index, Bessel coefficients. Phase-lock loop; limiter, pre-emphasis and de-emphasis, stereophonic FM broadcasting. Noise waveforms and characteristics. Cascade network, experimental determination of noise figure. Effect of noise on AM and FM systems. Block diagram of a superheterodyne AM radio receiver, AM broadcast band and specification. Antenna: TV broadcast band and specification. Signal format, transmitter and receiver block diagrams of Black and White TV, and Color TV.

30h (T); PR: CPE 311; C

CPE 471 Microprocessor System & Interfacing**3 Credits**

INTEL8086 microprocessor system: CPU, memory, I/O, and buses subsystems, basic operation of a microprocessor system, signal pinouts. Hardware & Software interrupts. Assembly language programming. Interfacing to memory, I/O devices, 8255 PPI, 8251USART, keyboard, keypad, serial LCD, ADC/DAC chips. Memory address decoding techniques, serial port & USB interfacing. Motorola MC68000 microprocessor system: CPU, memory, I/O, and buses subsystems, basic operation of a microprocessor system, signal pinouts. Assembly language programming. Interfacing to memory, I/O devices, 6850 ACIA, 6821 PIA USART, keyboard, keypad, serial LCD, ADC/DAC chips.
45h (T); PR: CPE321, CPE372; C

CPE 481 Laboratory Course III**2 Credits**

Laboratory experiments on Computer hardware and software, Electronics, communication and Assembly Language Programming
90h (P); C

CPE 492 Industrial Training II**12 Credits**

On the job experience in industry chosen for its relevance to student's major (24 weeks during 400 level rain semester and the long vacation after the semester).
270h (P); C

500 LEVEL COURSES**ABE 573 Engineer in Society****1 Credit**

Philosophy of Science and Engineering. History of Engineering and Technology, engineering societies. Engineers' code of conduct and ethics. Engineering and nation building-economy, politics, business, safety in Engineering and introduction in risk analysis invited lectures from professionals. 15h (T); C

BUL 506 Engineering Law**3 Credits**

Introduction and sources of law, formats of contracts, liabilities in torts: assaults, negligence and strict liability. Professional role and liabilities of engineers. Contract of Employment: independent contractors, workmen compensation, property law. Partnerships Intellectual property, copyright, trademarks and patent. Registration and incorporation of companies and effects. Case studies relating to professional issues, Arbitration.
45h (T); R

CPE 501 Design & Installation of Electrical & ICT Services**2 Credits**

Introduction to Health and safety at work act in Nigeria. Electrical safety and First aid. Electricity supply regulations. NCC and FCC Codes of Practice, Lighting and Illumination: Design and calculations for different situations and applications both domestic and industrial. Electrical Installation material selection and ratings: Cables, fittings, motors, generators, equipment. etc. Glare. Conduiting, Trucking and overhead Installation. Telecommunication Design & Installation: Telephone, PABX, cables, cablings, trucking, calculations, etc. Computer Networking: Design, Calculations, topology, cables, cabling, etc. Satellite and VSAT installation. Surge and lighting protections. Earthing: earth resistivity measurement, surge and lighting equipment selection and installation. Contract proposal and document preparation. Costing and preparation of BEME.
30 (T); C

CPE 502 Reliability and Maintainability of Computer Systems**2 Credits**

Introduction to reliability, maintainability, reliability specification and metrics. Application to computer hardware system, communication equipment, power systems, electronic components. Basic maintenance types, and procedures of computer and digital

communication system. Fault troubleshooting techniques. QoS and time of availability of data communication. Quality control techniques. Design for higher reliability, fault tolerance. Software Reliability: software reliability specification, software reliability Metrics, fault avoidance, fault tolerance, programming for reliability, software safety and hazard analysis. Comparison of hardware and software reliability. Software Quality and Assurance: definition of software quality, software quality factors, quality control, cost of quality, quality assurance. SQA activities, formal technical reviews, software quality metrics, statistical quality assurance. ISO Standards, Requirements and Certification
30h (T); C

CPE 531 Digital Signal Processing

3 Credits

Introduction: Advantages of digital over analogue signal processing, problems of digitization, overview of application of DSP, basic elements of DSP system. Digital Processing of analogue signals. Discrete Time Signals & systems: Discrete time sequences (signals), classification and determination of discrete time system, discrete time i/o description (difference equation), solution of difference equations, convolution, correlation, impulse response. z-transform. Discrete Fourier Analysis: Discrete Fourier Transform and properties, inverse DFT, truncated fourier transform, windowing, FFT algorithms. **Digital Filters.** Practical application of DSP in audio, and video.
45h (T); C

CPE532 Computer Security Techniques

2 Credits

History of cryptographic System, Public Key Systems, Digital Signature. Information Theory: Entropy, Perfect Secrecy, Unicity Distance, Complexity Theory, NP Completeness, Number Theory. Data Encryption Method Ciphers, Knapsack Ciphers, Breakable NP-Complete Knapsack, Encryption Standards DES, RSA, Elliptic Curves. Cryptographic Techniques: Block and Stream Ciphers, Autokey, Endpoints of Encryption, One-Way Ciphers, Password and Authentication, Secret Keys and Public Keys, Threshold Scheme. Video Scrambling techniques, Digital video encryption techniques: principle, IRDETO, Viaaccess, Videoguard, etc. Security and Legality Issues: Copyrights, Patents, Trade Secret, Ownership of Products, Computer Crimes, Ethnical Issue in Computer Security.
30h (T); C

CPE 541 Control Theory II

2 Credits

Definition of Root locus, Properties of root locus, sketching of root locus plots. Effect of open-loop zeros and poles. Root locus design concepts. **Frequency response analysis and design:** Bode diagram, Polar plot, Nichols plot. Nyquist stability criterion: non-mathematical description of Nyquist criterion, interpretation of stability. Relative stability - Gain and phase margins. Closed-loop frequency response analysis - M and N contours, Nichols chart. Compensation techniques: lag, lead and lag-lead compensation, PD, PI and PID controllers. Cascade compensation based on root-locus method. Introduction to Feedback compensation. Computer-aided design and analysis of control system.
30h (T); PR: CPE 441; C

CPE 543 Technopreneurship & CyberLaw

2 Credits

Creativity and Innovation. Trends in technology development. Entrepreneurship. **Business Plan Development. Financing business. Entrepreneurship case studies.** Discussion of Nigerian business environment, and illustrated with successful Nigerian entrepreneurs. **Intellectual property and copyright and licensing.** Corporate social responsibility.
30h (T); C

CPE 544: Introduction to Robotic & Automation

2 Credits

Robot classification and manipulation. Technology and history of development of robots. Applications. Direct and inverse kinematics: arm equation. Workspace analysis and trajectory planning. Differential motion and statics. Manipulator dynamics. End-of arm

tooling. Automation sensors. Robot vision. Work-cell support systems. Robot and system integration. Safety. Human interface. Robot control system. Circuit and system configuration. Task oriented control. Robot control programming. Fuzzy logic and AI based robot control. Fundamentals of automation. Strategies and economic consideration. Integration of systems. Impact to the production factory. Evaluation of conventional processes. Analysis of automated flow lines. Assembly systems and line balancing. Automated assembly systems. Numerical control and adaptive control. Robot applications.

30h (T); PR: CPE 441; E

CPE 545 Control Theory III

2 Credits

Digital control: Concept of sampling: Z-transform, inverse zero-order-hold, transfer functions of sampled data systems, stability analysis. Finite word length effect. Digital 3-term PID design. State Space: State variables of dynamic system, formulation of state vector differential equation, solution state equation, transition matrix, eigenvalues and eigenvectors. System response and stability. Nonlinear control: Common types and effects of nonlinearities, phase plane and describing function analysis, closed loop response, limit cycle, and stability. Introduction to Fuzzy control system

45h (T); PR: CPE 441; E

CPE 546 Embedded System and Design

2 Credits

Introduction to embedded system, components, characteristics, applications. Intel 8051/8031 Micro-controller: Features of the 8051/8031 family, block diagram and definitions of the pin of the 8051, I/O port structure, memory organization. Interfacing to external memory, keypad, seven-segment LED display, ADC and DAC chips, and input / output port expansion, description and uses of hardware development tools. MOTOROLA M6811 Micro-controller: Features of the M6811 family, block diagram and definitions of the pin of the M6811, I/O port structure, memory organisation: general purpose RAM, bit addressable RAM, register bank, special function registers, external memory, memory space mapping and decoding, bus control signals timing. Assembly Language Programming. On-chip peripheral devices and I/O interfacing. Introduction to PIC microcontrollers.

30h (T); PR: CPE 443; C

CPE 547 Power Electronics

2 Credits

Overview of Power Semiconductor Switches: Power diodes, Thyristors, Power MOSFET, G.T.O., IGBT, Field controlled switches (SiT and SiTH), Comparison of Semiconductor Switches, Desired Characteristics in Controllable Switches, Drive and Snubber Circuits. **Line-Commutated Diode Rectifiers:** Uncontrolled rectifier, Single-Phase Diode Bridge Rectifiers, Three-Phase Full-Bridge Rectifiers, Inrush Current and Over-voltages at Turn-On, Line-Current Harmonics and Power Factor, Phase-Controlled Rectifiers and Inverters. **DC-DC Switch-Mode Converters:** Basic Topologies, Buck converter, Boost converter, Buck-Boost Converter, Flyback Converter. **Switch-Mode DC-AC Inverters:** Pulse-Width Modulation, Single-Phase Inverters, Three-Phase Inverters, Effect of Blanking Time on Output Voltage in PWM Inverters. **Resonant Converters. Power Supply:** Switching Power Supplies, Electrical Isolation, Protection Circuits, Power Supply Specification, Power Line Disturbances, Power Conditioners, Uninterruptible Power Supplies.

30h (T); PR: CPE 321; E

CPE 548 Fuzzy Logic & Programming

2 Credits

Fuzzy set theory, set theoretic operations, comparison between crisp sets and fuzzy sets. Fuzzy operators. Fuzzy relations and compositions on the same and different product spaces. Max-Min composition, Max-Product composition, fuzzy relational matrix, sup-star composition. Hedges or modifiers of linguistic variables, fuzzy logic vs. probability. Fuzzy reasoning and implication, the fuzzy truth tables, traditional propositional logic and the rule of inference, the Modus Ponens and Modus Tollens, fuzzy modeling with causal IF-THEN statements. Fuzzy Models, fuzzy logic systems, combination of fuzzy basis functions,

universal approximator, fuzzy neural network, fuzzy associate memory matrix, self-learning fuzzy systems. Fuzzy logic system applications.

30h (T); E

CPE 549 Mobile Communication & Network

2 Credits

Evolution of mobile radio communications. Examples of mobile radio systems: radio paging, cordless telephones, cellular radio. A basic cellular system, Frequency reuse, Roaming, Hand-off strategies, Co-channel interference, Traffic and Grade of service, System capacity, Improving capacity of cellular system. Propagation path loss, multipath propagation problem, Raleigh fading, Rician distribution. Doppler effect. Field strength prediction models, co-channel interference and reduction, adjacent channel interference, near-far problem. Standards and overview of analogue and digital cellular systems: AMPS, TACS, GSM, CT2, PCN, DECT, PHS. Frequency management and channel assignment, multiple access techniques. GSM; CDMA; W-CDMA; UWC-136; Global Positioning System.

30h (T); E

CPE 551 PARALLEL PROCESSING

2 Credits

Parallelism in uniprocessors. Multiprocessing and Time-sharing. Parallel computer structures. Pipeline computers. Classification Scheme. Application of parallel processors. Trends towards Parallel computers. Principles of pipelining. Performance and cost. Instruction & Arithmetic Pipelining. Data dependencies. Internal forwarding. Multistreaming. Reservation tables & pipeline scheduling. CISC characteristics. RISC characteristics. RISC Vs CISC. RISC examples. RISC architectures. Design philosophy. Multiprocessor Architecture and Programming: Multiprocessor classification. Parallel Memory. Multiprocessor Operating Systems. Language features. Detection of parallelism in software. Multiprocessing Control. Data Flow Computers, VLSI computing structure.

30h (T); E

CPE552 Advanced Digital Design

2 Credits

Advanced features of VHDL (library, package and subprograms). Structural level modeling, Register-Transfer level modeling, FSM with datapath level modeling, Algorithmic level modeling. Introduction of ASIC, Types of ASIC, ASIC design process, Standard cell ASIC synthesis, FPGA Design Paradigm, FPGA synthesis, FPGA/CPLD Architectures. VHDL Design: Top-down design flow, Verification, simulation alternatives, simulation speed, Formal verification, Recommendations for verification, Writing RTL VHDL code for synthesis, top-down design with FPGA. VHDL synthesis, optimization and mapping, constraints, technology library, delay calculation, synthesis tool, synthesis directives. Computer-aided design of logic circuits.

30h (T); E

CPE 553 DIGITAL IMAGE PROCESSING

2 Credits

Introduction: definition, problems, and applications of digital image processing. Digital image acquisition devices. Digital image formats. Spatial filters, Edge detection techniques, segmentation methods. Image Morphology. Image enhancement. Image restoration techniques. Morphology. Fourier transform and Wavelet transform in image processing. Image registration techniques. Shape analysis. Image understanding. Artificial neural network and image understanding. Colour representation standards, equations, processing, quantization, and dithering. Case study: face recognition, fingerprint, iris, etc. Introduction to image compression techniques.

30h (T); E

CPE 554 Cryptography Principles & Applications

2 Credits

History of cryptographic System, Public Key Systems, Digital Signatures. Information Theory: Entropy, Perfect Secrecy, Unicity Distance, Complexity Theory, NP Completeness, Number Theory. Data Encryption Methods: Transposition Ciphers, Substitution Ciphers,

Product Ciphers, Exponentiation Ciphers, Knapsack Ciphers, Breakable NP-Complete Knapsack, Encryption Standards DES, RSA, Elliptic Curves. Cryptographic Techniques: Block and Stream Ciphers, Autokey, Endpoints of Encryption, One-way Ciphers, Password and Authentication, Secret Keys and Public Keys, Threshold Scheme. Video scrambling techniques. Digital video encryption techniques: principle, IRDETO, Viaaccess, Videoguard, etc. Security and Legality Issues: Copyrights, Patents, Trade Secret, Ownership of Products, Computer Crimes, Ethical Issue in Computer Security.

30h (T); E

CPE 556 Multimedia Technology & Programming

2 Credits

Definition of multimedia communication: asymmetry of content. Multimedia structures: Interactive TV, video-on-demand, video servers and set-up boxes. Virtual reality on the internet, virtual reality tool standards. Transmission techniques for business multimedia and kiosks. Design of multimedia services. Multimedia Programming: How pictures are encoded, how sound is encoded, text as unimedia. Techniques for multimedia coding: the MHEG model. Composition of multiple media. Referencing, Services, Technology and standards for broadcast-related multimedia. Digital video broadcasting standards, MPEG architecture, physical interfaces. Servers for interactive TV services. Overview of transmission systems: the ADSL concept, broadband multimedia delivery over copper, the hybrid fiber/copper concept.

30h (T); E

CPE 557 Digital Speech Processing

2 Credits

Mechanism of speech production and acoustic phonetics. Acoustic theory of speech production. Lossless tube models. Time-domain models for speech processing: short-time energy and average magnitude, speech versus silence discrimination using energy and zero-crossings, short-time average magnitude difference function, Pitch-period estimation. Digital representations of the speech waveform: Adaptive quantization, adaptive delta modulation, Differential PCM, PCM-to-ADPCM conversion. Short-time Fourier analysis: filter bank summation method of short-time synthesis. Design of digital filter banks using IIR and FIR filters. The complex cepstrum of speech. Format estimation. Linear predictive coding of speech.

30h (T); E

CPE 558 Mobile Applications Development

2 Credits

Mobile devices operating systems; Android, ios and windows mobile; Java apps; objective C apps. 30h (T); E

CPE 560 Electromagnetic Interference

2 Credits

Concern for electromagnetic compatibility, Circuit theory approach and field theory approach. Comparative analysis. **Coupling and Shielding:** Capacitive (electric) coupling. Inductive (magnetic) coupling. Shielding of cables. Balanced circuits. **Grounding:** Grounding of multiple-chassis systems. Signal ground connections. Safety ground connections. Layout and grounding of printed circuit board: layout consideration, current return path, power distribution within a PCB. **Radiation:** Radiation coupling between distant devices. Superposition of multiple electric and magnetic sources. Cabinet shielding. Absorption losses and reflection losses for nonmagnetic shields. Effects of shield apertures: current flow in shields, slot antenna theory, waveguide theory. Shield penetration by wires and cables. Interconnecting leads as antennas, treatment of power, low frequency and high frequency leads. **EMC Regulations and Measurements:** Civilian regulations. Measurement of radiated emissions. Anechoic Chamber. Test site calibration. Measurement of conducted emissions.

30h (T); E

CPE 561 Computer Organisation and Architecture**2 Credits**

Von-Neuman architecture vs Harvard architecture. Single address machine. Introduction to CISC and RISC architecture: principle of operation, merits, demerits. Storage and Input/Output Systems: Computer function (fetch and execute cycles), interrupts, interconnection structures (Bus structure and bus types), Overview of memory system, memory chip organization and error correction, cache memory, memory storage devices. Overview of I/O, programmed and interrupt-driven I/Os, DMA, I/O channel and I/O processor. Control Unit: Micro-operations, control of the CPU, hardwired implementation, control unit operation, micro-instruction sequencing and execution, micro-programmed control. Use INTEL family, and MOTOROLA family. Instruction Sets and Registers. Achieving high performance: pipelining, storage hierarchy, units with function dedicated for I/O. RISC processors. Operating System: Overview of operating systems.

45h (T); PR: CPE 336; C

CPE 593 Project I**4 Credits**

Original individual student project related to a prescribed Computer Engineering problem involving literature review, identification, definition and formulation of the problem, theoretical investigations, modelling simulation, analysis and design.

30h(T); 150h (P); C.

CPE 594 Project II**4 Credits**

Second phase of investigations involving the implementation of the designed model, debugging, calibration, testing, data collection and analysis, and presentation of a comprehensive written report of the investigations.

180h (P)

MME 524 Materials Selection and Economics**3 Credits**

Criteria for material selection. Material for structural application, and for electrical, electronics, chemical and nuclear applications. High temperature, creep resistant materials and materials for cryogenic applications. Corrosion resistant materials and materials for use in energy conversion systems. Fatigue-resistant materials. Materials for special applications, insulating materials, refractory and composite materials. Technical and economic considerations in materials selection – availability, durability, properties and cost. Material recycling: principles and economics. Economics of materials exploitation and usage with special regards to present and future availability.

45h (T); C.

OPTIONS

1. Embedded Systems & Automation
2. Computer Hardware & Software Systems

SUMMARY OF TAUGHT COURSES FOR B.ENG DEGREE PROGRAMME IN COMPUTER ENGINEERING**100 LEVEL****REQUIRED COURSES**

GNS 111 (2), GNS 112 (2), GNS 114 (1)

= 5 Credits**ELECTIVE COURSES:**

STA 131 (2), STA 124 (2)

= 4 Credits

At least 9 credits must be passed out of the following:

MAT 111(3), 113 (3), 112 (3), 114 (3)

= 9 Credits

At least 9 credits must be passed out of the following:

PHY 115 (2), 125 (3), 142 (2), 152 (3), 191 (1), 192 (1) = 9 Credits

At least 6 credits must be passed out of the following:

CHM 101 (3), 112 (2), 115 (2), 116 (1), 132 (2) = 6 Credits

Total = 5 Credits

200 LEVEL

COMPULSORY COURSES

CPE 283 (2), CPE 222 (6), CPE 284 (2) = 10 Credits

REQUIRED COURSES

ELE 201 (3), ABE 263 (3), CHE 241 (3), ELE275 (1), CVE 253 (3), GNS 211 (2), MEE 217 (2), MEE 235 (2), ABE 206 (2), CHE 242 (3), CHE 264 (3), ELE 202 (3), ELE 276 (2), CVE 254 (3), GNS 212 (2), MEE 218 (2), MEE 272 (2), GSE 202 (2) = 43 Credits

Total = 53 Credits

DE: GNS 111 (2), GNS 112 (2), GNS 114 (1) = 5 Credits

300 LEVEL

COMPULSORY COURSES

CPE 311 (3), CPE 321 (3), CPE 331 (3), CPE 341 (3), CPE 381 (1), CPE 312 (2), CPE 336 (3), CPE 342 (3), CPE 372 (3), CPE 382 (2), **CPE 392 (6)** = 32 Credits

REQUIRED COURSES

GNS 311 (2), GSE 301 (2), MEE 361 (3), MEE 362 (3), ABE 306 (2), ABE 376 (1) = 13 Credits

Total = 45 Credits

NOTE: Direct Entry (DE) students at 300 Level have additional required courses and these are **GNS 111 (2), GNS 112 (2), GNS 114 (1), GSE 202 (2), GNS 211 (2), and GNS 212 (2)** = 8 Credits

400 LEVELS

COMPULSORY COURSES

CPE 424 (2), CPE 436 (2), CPE 438 (2), CPE 444 (2), CPE 442 (3), CPE 452 (2), CPE 472 (3), CPE 482 (2), **CPE 492 (12)** = 30 Credits

REQUIRED COURSE

ABE 464 (2) = 2 Credits

Total = 32 Credits

500 LEVEL

COMPULSORY COURSES

CPE 501 (2), CPE 531 (3), CPE 543 (2), CPE 561 (2), CPE 593 (4), CPE 502 (2), CPE 532 (2), CPE 546 (2), CPE 594 (4), CPE 541 (2) = 25 Credits

REQUIRED COURSES

ABE 501 (3), ABE 573 (1), BUL 502 (3), MME 524 (3) = 10 Credits

ELECTIVE COURSES:

Students are expected to choose four of the courses in any of their preferred option, two in each semester.

Embedded Systems and Automation Option

CPE 545 (2), CPE 547 (2), CPE 544 (2), CPE 548 (2)

= 8 Credits

Computer Hardware and Software Systems

Only 8 credits must be passed out of the following courses in this option:

CPE 551 (2), CPE 553 (2), CPE 557 (2), CPE 552 (2),

CPE 554 (2), CPE 556 (2)

= 8 Credits

GRADUATION REQUIREMENTS

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| 1. Major Engineering Courses
(ABE, CHE, CPE, CVE, ELE, MEE, MME) | 128 Credits |
| 2. Students' Industrial Works Experience Scheme (SIWES) | 18 Credits |
| 3. Students Work Experience Programme (SWEP) | 6 Credits |
| 4. General Studies Courses: (GNS 111, 112, 114, 211, 212, 311) | 11 Credits |
| 5. Minimum Electives | 8 Credits |
| 6. Management, Law, Economics and Entrepreneurial Skill courses
(GSE 202 (2), GSE 301 (2), BUL 506 (3)) | 7 Credits |
| Total | = 178 Credits |

UTME: 178 Credits

DE (200L): 178 Credits

DE (300L): 131 Credits

COMPUTATION OF GRADE POINT

- 100 level courses are not used for computation of CGPA except GNS 111 (2), GNS 111 (2) and GNS 114 (1)
- The 18 credits of SIWES must be passed but they are not used for computation of CGPA
- The minimum Credits that will be used to compute the CGPA for all options are as follows:

Level	UTME	DE (200L)	DE (300L)
100 Level	5	-	-
200 Level	53	58	-
300 Level	39	39	50
400 Level	20	20	20
500 Level	43	43	43
Total	160 Credits	160 Credits	113 Credits

ACADEMIC REGULATIONS

Regulations guiding the conduct of students and staff are as stated in the University Undergraduate Academic Programme and the Staff Condition of Service book respectively.